

Gregory M. Campbell, PhD
CV - September 2026

✉ gregory.campbell@scranton.edu
in/gregorymcampbell
🌐 gmcampbell.github.io

Teaching Experience

Assistant Professor of Mechanical Engineering - University of Scranton **Fall 2026-Present**

- Fall 2026 courses: Computer Aided Design I, Computational Tools for Physics and Engineering Lab

Visiting Assistant Professor of Mechanical Engineering - Lafayette College **Fall 2025 & Spring 2026**

- Fall 2025 courses: Design and Manufacturing; Strength of Materials
- Spring 2026 courses: Dynamics; Instrumentation and Data Acquisition Lab

Teaching Assistant (TA) - Engineering Mechanics: Dynamics at University of Pennsylvania **Spring 2022**

- Prepared and presented weekly recitation lectures to ~20 student class in accordance with course goals

Teaching Assistant (TA) - Design of Mechatronic Systems at University of Pennsylvania **Spring & Fall 2021**

- Prepared and presented weekly recitation lectures in remote & hybrid format to class of 100+ students
- Assisted 100+ students in fabrication and coding of teleoperated and autonomous mobile robots
- Nominated for Spring 2021 Outstanding TA Award

Education

University of Pennsylvania, Philadelphia, PA **2019-2025**

PhD in Mechanical Engineering and Applied Mechanics

- Conducted soft robotics research in ModLab, Pikul Research Group, and GRASP
- PhD thesis: ‘Elastomeric strain limitation for design of soft pneumatic actuators’

Stanford University, Stanford, CA **Class of 2016**

Master of Science in Mechanical Engineering, with depth in Mechatronics

- Received Graduate Engineering Fellowship Award to study and research in Mechanical Engineering

Villanova University, Villanova, PA **Class of 2014**

Bachelor of Science in Mechanical Engineering

- Concentration in Dynamic Systems, with Minors in Mechatronics and Honors
- Received Mechanical Engineering Outstanding Student Award (top Mechanical Engineering graduate)
- Received Robert D. Lynch Award for academic achievement and service in Engineering

Additional Work Experience

Senior Associate - Vehicle Engineering Consultant at Exponent, Inc., Philadelphia, PA **2016-2019**

- Performed accident analysis and reconstruction on dozens of incidents for various clients
- Accessed electronic modules to extract accident relevant data from passenger and heavy vehicles

R&D Advanced Surgical Technologies Intern at Medtronic, Boulder, CO **Summer 2015**

- Implemented, through Arduino to C++ Serial Communication, a controller for wristed actuation
- Developed, through CAD and rapid prototyping, 4 iterations of a future surgical device

“RISE” Research Intern at Technische Universität Kaiserslautern, Germany **Summer 2013**

- Investigated, through seat experimentation, the effects of whole body vibrations on tractor drivers

Publications

- Campbell, G.M.**, Cao, Y., Escritor, H., Zhou, Z., & Yim, M. (2026). Co-Optimization of Design and Manufacturing Parameters for Low-Cost Robotic Actuation. *International Symposium on System Integration (SSI) 2026*.
- Campbell, G. M.**, Muhaxheri, G., Guilhoto, L.F., Santangelo, C.D., Perdikaris, P, Pikul, J., & Yim, M. (2025). Active Learning Design: Modeling Force Output for Axisymmetric Soft Pneumatic Actuators. *Oct. 2025 IEEE Robotics and Automation Letters*, vol. 10, no. 10, pp. 10838-10845.
- Campbell, G. M.**, Roosa, R.D., Turner, K., Pikul, J., & Yim, M. (2024). Control of Silicone-Sheathed Electrostatic Clutches for Soft Pneumatic Actuator Position Control. *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*
- Levine, D. J., Lee, O. A., **Campbell, G.M.**, McBride, M. K., Kim, H. J., Turner, K. T., Hayward, R.C., & Pikul, J. H. (2023). A Low-Voltage, High-Force Capacity Electroadhesive Clutch Based on Ionoelastomer Heterojunctions. *Advanced Materials* 35(46), 2304455.
- Campbell, G.M.**, Yin, J., Song, Y., Gandhi, U., Yim, M., & Pikul, J. (2022). Electroadhesive Clutches for Programmable Shape Morphing of Soft Actuators. *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 11594-11599.
- Yin, J., **Campbell, G. M.**, Pikul, J., & Yim, M. (2022). Multimodal proximity and visuotactile sensing with a selectively transmissive soft membrane. *2022 IEEE 5th International Conference on Soft Robotics (RoboSoft)*, pp. 802-808.
Winner of RoboSoft 2022 Best Student Paper Award.
- Skiera, J., Crosby, C., Bare, C., Paradiso, M., & **Campbell, G.** (2019). Passenger Vehicle Dynamic Response and Characterization of Side Structure during Low-to Moderate-Speed Side Impacts. *SAE International Journal of Advances and Current Practices in Mobility*, 1(2019-01-0420), pp. 918-950.
- Brugarolas, T., Gianola, D. S., Zhang, L., **Campbell, G.M.**, Bassani, J. L., Feng, G., & Lee, D. (2014). Tailoring and understanding the mechanical properties of nanoparticle-shelled bubbles. *ACS applied materials & interfaces*, 6(14), pp. 11558-11572.

Patent Application

Pikul, J., **Campbell, G.M.**, U.S. Provisional Application No.: 63/344,248 (May 20, 2022), Programmable Soft Actuator

Presentations & Workshops (Soft Robotics)

- Campbell, G. M.**, Muhaxheri, G., Guilhoto, L.F., Santangelo, C.D., Perdikaris, P, Pikul, J., & Yim, M. (2026). Active Learning Design: Modeling Force Output for Axisymmetric Soft Pneumatic Actuators. Poster presentation of the R-AL publication of the same name. 2026 International Conference on Robotics and Automation (ICRA 2026). Vienna, Austria.
- Campbell, G.M.** (2024). Don't Just Guess... Gauss: Basics of Gaussian Process Regression for Autonomous Experimentation. Presentation as part of the University of Pennsylvania - Universidad de Puerto Rico en Humacao Partnership for Research and Education in Materials Seminar Series on Research in Autonomous Experimentation. November 2, 2023. Virtual.
- Yin, J., **Campbell, G.M.**, Pikul, J., & Yim, M. (2022). Multimodal Proximity and Visuotactile Sensing With a Selectively Transmissive Soft Membrane. Poster & presentation at: 39th IEEE International Conference on Robotics and Automation (ICRA 2022) "Leveraging Advancements in Smart Materials Science: Soft Robots Gaining New Abilities through Smart and Functional Materials" Workshop. May 27, 2022. Philadelphia, PA.
- Campbell, G.M.**, Yin, J., Yim, M., & Pikul, J. (2022). Electroadhesive Clutches for Programmable Shape Morphing of Soft Actuators. Poster & demonstration at: 39th IEEE Conference on Robotics and Automation (ICRA 2022) "Determining Appropriate Metrics and Test Methods for Soft Actuators in Robotic Systems" Workshop. May 23, 2022. Philadelphia.
- Kim, K. E., Yin, J., **Campbell, G.**, Pikul, J., & Santangelo, C. (2022). Contact force estimation from deformation of surface. Bulletin of the American Physical Society. Presentation at: APS March Meeting 2022.

Presentations & Workshops (Pedagogical)

Campbell, G.M., & Sklaroff E.M. (2024). Alternatives to High Stakes Assessments. Part of the Center for Excellence in Teaching, Learning and Innovation (CETLI) University-Wide Graduate Workshops Series. March 14, 2024. Philadelphia, PA.

Campbell, G.M. (2023). How To Solicit & Engage with Student Feedback. Teaching assistant training follow-up workshop. October 2, 2023. Philadelphia, PA.

Convened and facilitated Mechanical Engineering faculty & staff-led workshops through UPenn's Center for Excellence in Teaching and Innovation:

- Kevin Turner (2024), Lessons Learned from Implementing Structured Active Learning for an Undergrad Course
- Mark Yim (2024), Intentional Ineptitude: Promoting Self-learning in Engineering
- Dustyn Roberts (2024), Indistinguishable from magic: Teaching advanced technology with real-world applications
- Devin Carroll (2024), Lessons Learned in Lectureship: the Road from Grad Student to Lecturer
- Bruce Kothmann (2023), Learning by Mistake: Encouraging a Growth Mindset
- Laura Stubbs (2023), Inclusive Teaching Lessons from my Time in the Military and Beyond
- Igor Bargatin (2023), Attendance Matters: Cultivating Classroom Commitment

Fellowships, Certificates, Honors, and Awards

NSF NRT Soft Autonomous Experimentation (AE) Fellow	2023-2025
UPenn Center for Excellence in Teaching, Learning and Innovation (CETLI) Graduate Fellow	2023-2024
UPenn CETLI Certificate in College and University Teaching	2023
RoboSoft Best Student Paper Award	2022
Stanford Graduate Engineering Fellowship	2015-2016
Villanova Mechanical Engineering (ME) Outstanding Student Award (top ME graduate)	2014
Villanova Robert D. Lynch Award for academic achievement and service in Engineering	2014
Barry Goldwater Scholarship Honorable Mention	2014

Service Activities

Mentoring UPenn engineering students in undergraduate research (see below)	2021-2025
Topics: conductive elastomers, pneumatic actuator design, biomechanical simulation, mobile robotics	
GRASP Lab social committee member	2023-2025
Army Educational Outreach Program (AEOP) mentor for high school students	Summer 2023
Student's research was recognized with 1st prize in regional JSHS poster competition	
Volunteer coordinator at International Conference on Robots and Automation (ICRA 2022)	May 2022
ModLab social organizer	2020-2023
Mechanical Engineering Graduate Association (MEGA) board member	2020-2021
Judge for high school student projects at the Science Olympiad at UPenn	2020-2022

Teachable Skills

Software

Proficient in Python, Matlab, C, C++ (Arduino), SolidWorks, Autodesk Fusion, and $\text{\LaTeX}$

Machine Shop and Rapid Prototyping

Experienced with 3D printers, laser cutters, elastomer casting, manual milling machine, vertical band saw, etc.

Student Research Mentoring

Hannah, Yi, & Zihao

Spring & Summer 2025

- Project: Active learning (Bayesian Optimization) for rapid prototyping design
- Outcome: Authorship on published paper presented at international conference

Jason

Summer & Fall 2024

- Project: Soft membrane fabrication and experimentation with active learning
- Outcome: Dataset used in paper accepted to international journal (acknowledgement in paper)

Brennan & Tina

Summer 2023

- Project: Soft membrane array for object manipulation
- Outcome: 1st prize in regional Junior Science and Humanities Symposium (JSHS) poster competition

Charles

Summer 2023

- Project: Soft robotic seat lift from inextensible materials
- Outcome: Working prototype with proof-of-concept video

Kayla, Ashna, & Theo

Summer 2022

- Project: ‘Vine-crawler’ robots for local strain limitation on soft robots
- Outcome: Reproduction of MIT Rovables platform

Eric

Spring 2022

- Project: 2D motion tracking with computer vision using Robot Operating System (ROS)
- Outcome: Working prototype and instructions for reproduction

Sharon

Spring & Summer 2021

- Project: Biomechanical simulation (OpenSim) of sit-to-stand for atrophied patient
- Outcome: Simulation of sit-to-stand using OpenSim through MATLAB

Tate

Fall 2019

- Project: Embedded electrode for electroadhesion of soft membranes
- Outcome: Basic prototype and experimental data